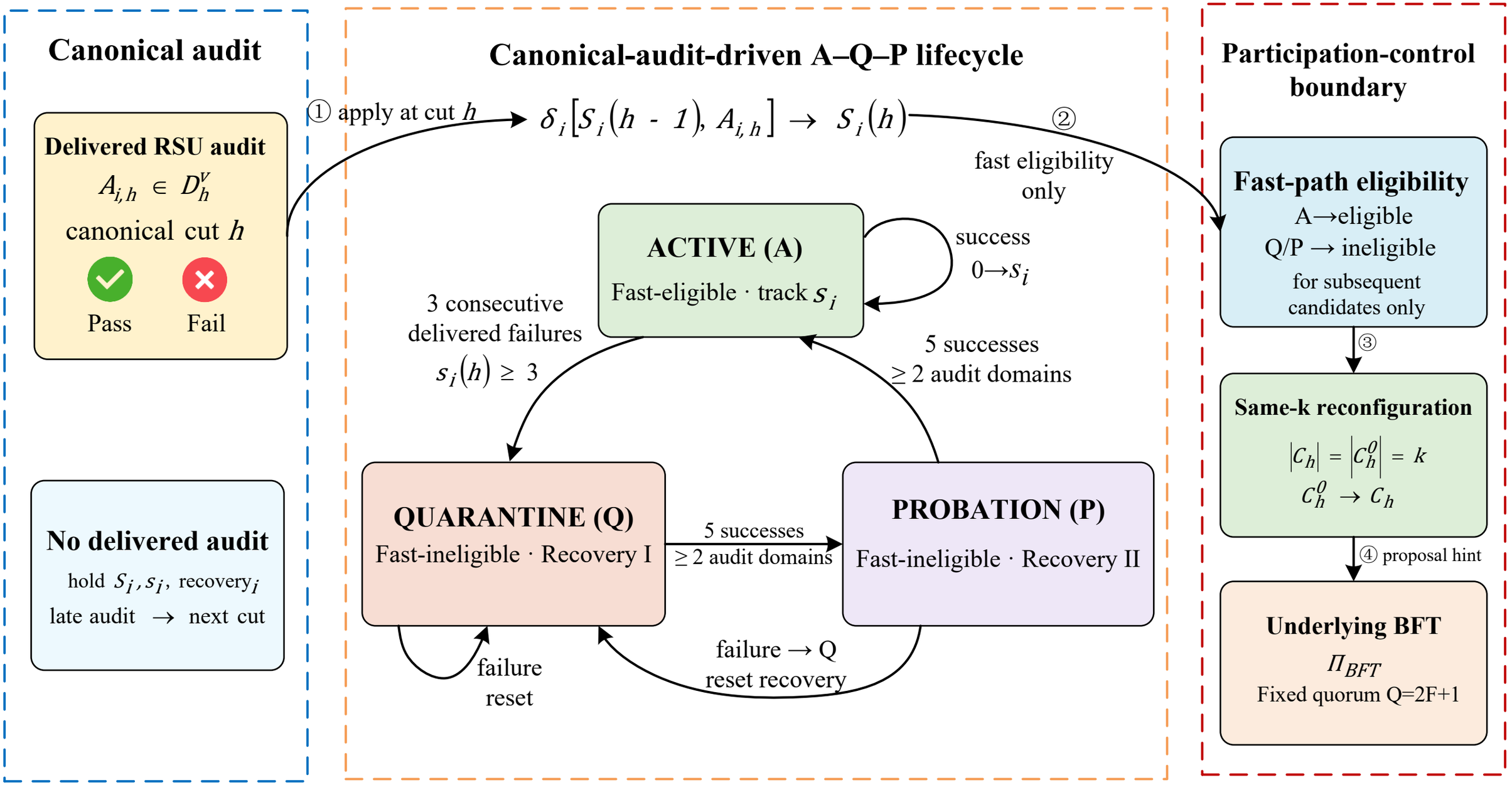

Canonical audit
Delivered RSU audit
$A_{i,h} \in D_h^v$
canonical cut h
Pass
Fail
No delivered audit
hold S_i, s_i, recovery$_i$
late audit → next cut
Canonical-audit-driven A–Q–P lifecycle
① apply at cut h
$\delta_i[S_i(h-1), A_{i,h}] \to S_i(h)$
② fast eligibility only
ACTIVE (A)
Fast-eligible · track s_i
success $0 \to s_i$
3 consecutive delivered failures $s_i(h) \geq 3$
5 successes ≥ 2 audit domains
QUARANTINE (Q)
Fast-ineligible · Recovery I
5 successes ≥ 2 audit domains
PROBATION (P)
Fast-ineligible · Recovery II
failure reset
failure → Q reset recovery
Participation-control boundary
Fast-path eligibility
A→eligible
Q/P → ineligible
for subsequent candidates only
③
Same-k reconfiguration
$|C_h| = |C_h^0| = k$
$C_h^0 \to C_h$
④ proposal hint
Underlying BFT
Π_{BFT}
Fixed quorum Q=2F+1